

The Gravity Equation

From Newton to Anderson and van Wincoop

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Section 1

Origins

Nature and Nature's laws

“Nature and Nature's laws lay hid in night. God said, Let Newton be! and all was light.”

— Alexander Pope

Newton (1687): a body i attracts a body j with a force proportional to the product of their masses and inversely proportional to the square of the distance.

$$X_{ij} = G \frac{Y_i Y_j}{\tau_{ij}^2}$$

- In trade, the masses are **GDPs** and X_{ij} is the flow of goods from i to j .
- Tinbergen (1962) is the first to test this relation — with success, and with no theory behind it.

The world is flat?

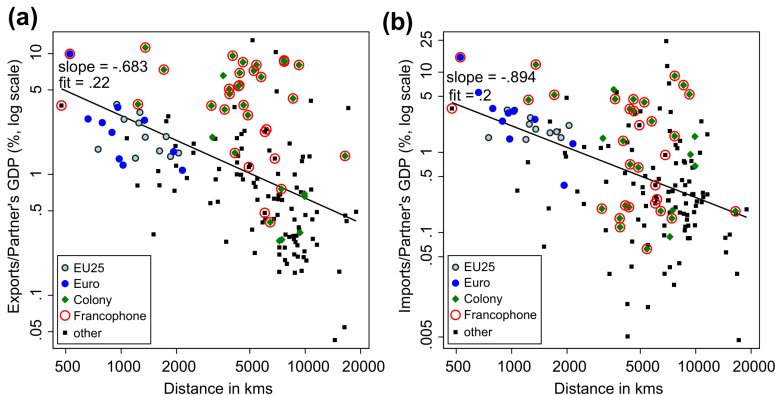
- The claim: transport and communication costs have collapsed, so **geography no longer matters**.
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The data disagree. Trade falls with distance, with an elasticity close to -1 .

Distance still matters



France's exports (a) and imports (b), 2006, relative to the partner's GDP, against distance. Both axes in logs. Source: Head and Mayer (2014).

From intuition to theory

The Newtonian equation is an **analogy**, not a model. Three consequences.

- It says nothing about the **mechanisms** behind the flows: why do consumers buy from abroad, and how do prices adjust?
- It cannot be used for **counterfactuals**. We can predict only “all else equal” — and in general equilibrium nothing else is equal.
- It has **no parameters we can interpret**. What is G ? What does the exponent 2 on distance mean for welfare?

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Where we are going

We now derive the same equation from a demand system. The theory adds two terms that the analogy misses, and those two terms change everything.

Section 2

Anderson and van Wincoop

The setup, in three assumptions

Anderson and Van Wincoop (2003) — AvW hereafter.

- 1 **Complete specialisation** (Armington (1969)): each country produces one variety. Goods are differentiated by their **origin**.
- 2 **CES preferences**, identical across countries, with elasticity of substitution $\sigma > 1$:

$$U_j = \left(\sum_{i=1}^R \beta_i^{\frac{1-\sigma}{\sigma}} c_{ij}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

- 3 **Iceberg trade costs** $\tau_{ij} \geq 1$: a good sold at p_i at home costs $p_{ij} = p_i \tau_{ij}$ in j .
 β_i is a taste shifter for goods from i ; E_j is expenditure in j .

Step 1 — Demand

Maximising U_j subject to $\sum_i p_{ij}c_{ij} = E_j$ and using $X_{ij} = p_{ij}c_{ij}$:

$$X_{ij} = (\beta_i p_i \tau_{ij})^{1-\sigma} \frac{E_j}{P_j^{1-\sigma}}$$

where P_j is the CES price index in j :

$$P_j = \left[\sum_{i=1}^R (\beta_i p_i \tau_{ij})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

- Trade rises with the buyer's expenditure E_j and falls with the bilateral cost τ_{ij} .
- P_j measures how easily consumers in j reach **all** suppliers. AvW call it the **inward multilateral resistance**.

Step 2 — Market clearing

Country i sells its whole production, at home and abroad. Summing demand over all destinations:

$$Y_i = \sum_{j=1}^R X_{ij} = (\beta_i p_i)^{1-\sigma} \sum_{j=1}^R \tau_{ij}^{1-\sigma} \frac{E_j}{p_j^{1-\sigma}}$$

Solve for the unobserved factory price:

$$(\beta_i p_i)^{1-\sigma} = \frac{Y_i}{\Omega_i^{1-\sigma}}, \quad \Omega_i = \left[\sum_{j=1}^R \frac{E_j}{p_j^{1-\sigma}} \tau_{ij}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Ω_i measures how easily producers in i reach **all** markets: the **outward multilateral resistance**.

Step 3 — The structural gravity equation

Substituting back into demand:

Structural gravity

$$X_{ij} = \underbrace{\frac{Y_i}{\Omega_i^{1-\sigma}}}_{\text{exporter}} \underbrace{\frac{E_j}{P_j^{1-\sigma}}}_{\text{importer}} \underbrace{\tau_{ij}^{1-\sigma}}_{\text{bilateral}}$$

with the two resistances defined by each other:

$$P_j^{1-\sigma} = \sum_{i=1}^R \frac{Y_i}{\Omega_i^{1-\sigma}} \tau_{ij}^{1-\sigma}$$

- Same shape as Newton — two masses, one distance term.
- But the masses are now **deflated by the resistances**, and those depend on the trade costs of *every* country pair.

What the resistances do

- $\tau_{ij}^{1-\sigma}$ is not a cost but a measure of **openness**: since $\sigma > 1$, it rises when τ_{ij} falls. It runs from 0 (autarky) to 1 (free trade).
- The system (P_j, Ω_i) has **no analytical solution**: each price index depends on all the others.

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Third-country effect

A rise in trade costs *anywhere* changes P_j and Ω_i , and therefore changes X_{ij} — even when τ_{ij} itself has not moved.

Two countries that sign nothing can see their trade change because others signed.

Two results worth remembering

Theory. Small countries gain most from opening and lose most from closing. They import relatively more, so their price index is more sensitive to trade costs. Behind results such as Mexico gaining most from NAFTA (Caliendo and Parro 2014) or the UK losing most from Brexit (Dhingra et al. 2017).

Empirics. If the resistances are omitted from the regression, they end up in the error term. They are correlated with τ_{ij} , so every bilateral coefficient is biased — an omitted variable problem, not a nuisance.

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This is why the naive equation gets policy effects wrong. We now look at three famous cases.

Section 3

Taking it to the data

Three puzzles

Estimating the naive equation in logs, $\ln X_{ijt} = \ln Y_{it} + \ln Y_{jt} + \ln \tau_{ij} + \epsilon_{ijt}$, produced three results that made no sense.

- ❶ **The GATT/WTO puzzle.** Rose (2004): members do not trade more than non-members.
- ❷ **The currency union puzzle.** Rose (2000): a common currency *triples* trade. Far too large to be credible.
- ❸ **The border puzzle.** McCallum (1995): Canadian provinces trade **22 times** more with each other than with US states, despite NAFTA.

Three resolutions

- ① **GATT/WTO.** Larch et al. (2025) re-estimate with exporter-time and importer-time fixed effects, pair fixed effects, and **domestic trade flows**. Membership raises trade between members by about 171%. The old result was an artefact of a missing multilateral resistance.
- ② **Currency unions.** Glick and Rose (2002) shrink the effect to a doubling; later work shrinks it further. The unresolved issue is **selection**: countries never sign at random.
- ③ **Borders.** Anderson and Van Wincoop (2003) show the effect falls from 22 to **10.7** once the general equilibrium constraints are imposed. Their paper *is* the resolution of the puzzle.

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In all three cases the fix is the same: put the resistances back in.

Fixed effects

In a panel, the theory tells us exactly what to control for:

$$\ln X_{ijt} = f_{it} + f_{jt} + \kappa \tau_{ijt} + \epsilon_{ijt}$$

- f_{it} absorbs $\ln(Y_{it}/\Omega_{it}^{1-\sigma})$ — the exporter's mass **and** outward resistance.
- f_{jt} absorbs $\ln(E_{jt}/P_{jt}^{1-\sigma})$ — the importer's mass **and** inward resistance.
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The price of consistency: GDPs can no longer be estimated, because they are collinear with the fixed effects. The theory says they belong *inside* f_{it} and f_{jt} .

Section 4

Exercise

Building the data from scratch

Estimating gravity is easy once someone has assembled the data. Doing the assembly yourself is the part that teaches you something (don't do that with an AI).

Four files, three of them to be merged before anything can be estimated:

Trade flows	bilateral exports, by year
GDP	one value per country and year
Distances	bilateral, plus contiguity, language, colonial links
Correspondence	the table that lets the others speak to each other

Two files that cannot be merged

t	v	i	j	hs6_bis
1998	1.997	12	24	848190
2000	1.977	12	24	271000
2002	29.869	12	24	730420
2003	2.842	12	24	271000
2004	4.975	12	24	271000
2006	10.15	12	24	271000
2006	13.642	12	24	901580
2007	31.208	12	24	271000
2009	2.945	12	24	070952
2009	10.443	12	24	271000
2009	2.779	12	24	400950
2009	22.66	12	24	481820
2009	19.619	12	24	730799
2009	873.449	12	24	730840
2009	4.856	12	24	841320
2009	6.172	12	24	841381
2009	47.979	12	24	841391
2009	4.019	12	24	848180
2009	173.47	12	24	851539
2009	64.362	12	24	960900
2009	52.342	12	24	901590
2010	4.765	12	24	321490
2010	1.303	12	24	482230

isoi	isoj	contig	colony	dist
ABW/	ABW/	0	0	5.225315
ABW/	AFG	0	0	13257.81
ABW/	AGO	0	0	9516.913
ABW/	ALA	0	0	983.2682
ABW/	ALB	0	0	9091.742
ABW/	AND	0	0	7572.788
ABW/	ANT	0	0	136.3848
ABW/	ARE	0	0	12735.01
ABW/	ARG	0	0	5396.22
ABW/	ARM	0	0	11107.78
ABW/	ATG	0	0	1024.995
ABW/	AUS	0	0	15237.5
ABW/	AUT	0	0	8719.13
ABW/	AZE	0	0	11503
ABW/	BDI	0	0	11120.47
ABW/	BEL	0	0	7847.07
ABW/	BEN	0	0	7979.945
ABW/	BFA	0	0	7414.016
ABW/	BGD	0	0	15473.46
ABW/	BGR	0	0	9348.039
ABW/	BHR	0	0	12308.36
ABW/	BHS	0	0	1588.515
ABW/	BIH	0	0	8936.664

Trade flows — i and j are **numbers**

Distances — countries are **letters**

The merge key simply does not exist. A fourth file, the **correspondence table**, maps one coding onto the other — and it is itself incomplete.

What you should end up with

i	j	t	_TYPE_	_FREQ_	v	isoj	countryj	isoi	contig	colony	dist	countryi	yi	yj	lyi	lyj	ldist	ltrade
894	12	1993	0	1	7033.163	DZA	Algeria	ZMB	0	0	6396.462	Zambia	3.27E+09	4.99E+10	21.90913	24.63422	8.7635	15.76615
894	12	2000	0	1	5.033	DZA	Algeria	ZMB	0	0	6396.462	Zambia	3.24E+09	5.48E+10	21.89813	24.72677	8.7635	8.523772
894	12	2002	0	1	14.548	DZA	Algeria	ZMB	0	0	6396.462	Zambia	3.72E+09	5.71E+10	22.03594	24.76725	8.7635	9.585209
894	12	2004	0	3	5.43	DZA	Algeria	ZMB	0	0	6396.462	Zambia	5.42E+09	8.5E+10	22.41395	25.16608	8.7635	8.599694
894	12	2005	0	2	13.764	DZA	Algeria	ZMB	0	0	6396.462	Zambia	7.16E+09	1.02E+11	22.69134	25.34824	8.7635	9.529812
894	12	2006	0	1	22.188	DZA	Algeria	ZMB	0	0	6396.462	Zambia	1.07E+10	1.16E+11	23.0912	25.47686	8.7635	10.00731
894	12	2007	0	2	47.69	DZA	Algeria	ZMB	0	0	6396.462	Zambia	1.14E+10	1.34E+11	23.15776	25.62111	8.7635	10.77248
894	12	2008	0	1	11.231	DZA	Algeria	ZMB	0	0	6396.462	Zambia	1.43E+10	1.74E+11	23.3845	25.88232	8.7635	9.326433
894	24	1995	0	8	326.855	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.48E+09	5.04E+09	21.96962	22.34058	7.490988	12.69727
894	24	1997	0	26	1048.26	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.91E+09	7.68E+09	22.0869	22.76129	7.490988	13.86264
894	24	1998	0	20	549.728	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.24E+09	6.45E+09	21.89798	22.5866	7.490988	13.21718
894	24	1999	0	7	184.962	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.13E+09	6.15E+09	21.86473	22.54045	7.490988	12.12791
894	24	2000	0	12	309.774	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.24E+09	9.13E+09	21.89813	22.93474	7.490988	12.6436
894	24	2001	0	6	22.583	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.64E+09	8.94E+09	22.01441	22.91336	7.490988	10.02495
894	24	2002	0	8	406.793	AGO	Angola	ZMB	1	0	1791.822	Zambia	3.72E+09	1.14E+10	22.03594	23.15966	7.490988	12.91606
894	24	2003	0	17	1048.287	AGO	Angola	ZMB	1	0	1791.822	Zambia	4.37E+09	1.4E+10	22.19891	23.35919	7.490988	13.86267
894	24	2004	0	23	393.519	AGO	Angola	ZMB	1	0	1791.822	Zambia	5.42E+09	1.98E+10	22.41395	23.7077	7.490988	12.88288
894	24	2005	0	19	555.879	AGO	Angola	ZMB	1	0	1791.822	Zambia	7.16E+09	3.06E+10	22.69134	24.14532	7.490988	13.22831
894	24	2006	0	18	362.422	AGO	Angola	ZMB	1	0	1791.822	Zambia	1.07E+10	4.52E+10	23.0912	24.53355	7.490988	12.80056
894	24	2007	0	21	565.444	AGO	Angola	ZMB	1	0	1791.822	Zambia	1.14E+10	5.93E+10	23.15776	24.80526	7.490988	13.24537
894	24	2008	0	43	1289.234	AGO	Angola	ZMB	1	0	1791.822	Zambia	1.43E+10	8.34E+10	23.3845	25.14672	7.490988	14.06956
894	108	1993	0	15	4578.72	BDI	Burundi	ZMB	0	0	1351.626	Zambia	3.27E+09	9.39E+08	21.90913	20.65993	7.209064	15.33693
894	108	1994	0	14	6507.88	BDI	Burundi	ZMB	0	0	1351.626	Zambia	3.35E+09	9.25E+08	21.9312	20.64534	7.209064	15.68852
894	108	1995	0	15	7970.025	BDI	Burundi	ZMB	0	0	1351.626	Zambia	3.48E+09	1E+09	21.96962	20.72369	7.209064	15.8912
894	108	1996	0	8	3699.062	BDI	Burundi	ZMB	0	0	1351.626	Zambia	3.27E+09	8.69E+08	21.90815	20.58289	7.209064	15.12359
894	108	1997	0	27	5796.005	BDI	Burundi	ZMB	0	0	1351.626	Zambia	3.91E+09	9.73E+08	22.0869	20.69579	7.209064	15.57268

One row per exporter–importer–year, carrying trade, distance, both GDPs and the bilateral dummies — plus the logs you will regress.

Three difficulties (at least)

- ❶ **Country codes do not match.** Countries are numbers in one file and letters in another. You need the correspondence table — and it is incomplete.
- ❷ **Names change over time.** Zaire becomes the Democratic Republic of the Congo, Yugoslavia becomes Serbia and Montenegro. A naive merge silently drops them.
- ❸ **GDP must be merged twice,** once on the exporter and once on the importer. Rename carefully, or you will overwrite the first merge with the second.

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Losing observations in a merge is not a technical detail. It is a sample selection decision, and it should be a deliberate one.

Your task

- ➊ Merge the trade, distance and GDP files into a single bilateral panel.
- ➋ Estimate the gravity equation with a **two-way fixed effects** regression:

$$\ln X_{ijt} = \alpha + \beta_1 \ln Y_{it} + \beta_2 \ln Y_{jt} + \beta_3 \ln \text{dist}_{ij} + f_i + f_j + f_t + \varepsilon_{ijt}$$

- ➌ Comment on the distance elasticity. Compare it with the slope on the figure we saw at the start, and with the value the literature reports.

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- 3 Comment on the distance elasticity. Compare it with the slope on the figure we saw at the start, and with the value the literature reports.

The full derivation, the data and the corrections are in the written course.

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